COURSE INFORMATION

School/Faculty:	Faculty of Computing		
Program name:	BSc in Software Engineering with honours		
Course code:	SCSR2043	**Academic Session/Semester:**	20252026/1
Course name:	Operating Systems	**Pre/co requisite** (*course name and code, if applicable*):	
Credit hours:	3		

Course synopsis	This course **provides** a comprehensive introduction to operating systems as the essential interface between computer hardware and users. It covers fundamental concepts, structures, and functions of operating systems, including process management, concurrency and synchronization, deadlocks, memory and virtual memory management, file systems, secondary storage, and input/output management. By integrating these concepts, students gain a solid understanding of how operating systems are designed and how they coordinate resources to ensure system efficiency. The course employs **lectures, guided labs, and project-based learning**, allowing students to apply theoretical knowledge to practical scenarios. By the end of the course, students will be able to **explain** and **analyze** core OS mechanisms, **distinguish and evaluate** system processes, and demonstrate **collaborative** problem solving, **critical thinking**, and effective **communication** in addressing real-world system challenges.
Course coordinator (*If applicable*)	Farkhana Binti Muchtar, Dr.

Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail** *@utm.my*
	Farkhana Binti Muchtar, Dr.	N28-439-09		farkhana
	Mohd Kufaisal bin Mohd Sidik, Dr.	N28-346-05		kufaisal

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	*** CLO**	**PLO ** (MQF Cluster Code)**	***** Taxonomies and **** Graduate Attributes**	**T&L methods**	*******Assessment methods**
CLO1	Apply operating system knowledge to solve problems by employing integrated **problem-solving** skills.	PLO1 (C1)	C3 SC1	Lectures	T, F
CLO2	Evaluate programs for Operating System–related operations or services using **critical thinking**.	PLO2 (C2)	C5 SC2	Lectures Hands-on Lab	L, F
CLO3	Demonstrate **collaborative** leadership in a team by taking initiative, guiding peers, respecting diverse perspectives, and coordinating efforts to achieve a common goal.	PLO8 (C3F)	A3 A(CC2)	Project-based learning	PR, Pr

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.
* Up to 5 CLO
Refer *** Taxonomies of Learning and **** UTM's Graduate Attributes for UG and Generic Skills for PG, where applicable for measurement of outcomes achievement
***** T – Test; Q – Quiz; HW – Homework; Asg – Assignment; PR – Project; Pr – Presentation; F – Final Exam etc.

** **MQF Cluster Code**
C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numeracy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism

Details on Innovative T&L practices:

No.	Type	Implementation
1	Hands-on Lab	Conducted through guided in-lab sessions where students apply theoretical knowledge using practical tools, coding exercises, or simulations.
2	Project-based Learning	Conducted through a given set of case study. Students in a group of 3 (max.) are required to create animation / simulation of algorithms in the case study with relation to the related techniques introduced in the course.

Transferable skills (*generic skills learning in course of study which can be useful and utilised in other settings*) :

This course enables students to build transferable skills that extend beyond technical knowledge of operating systems. By applying **problem-solving** and **critical thinking**, they learn to analyze challenges and make sound decisions. Team projects emphasize **collaboration** and **communication**, preparing students to contribute effectively while also taking initiative, guiding peers, and motivating groups toward shared goals. These experiences strengthen students' **leadership skills**, equipping them to adapt and apply their knowledge confidently in diverse professional contexts.

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO*	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			
Week 1	Introduction 1	CLO1	2	1								2	5
Week 2	Introduction 2	CLO1	2	1							1	2	6
Week 3	OS Structures	CLO1	2	1								2	5
Week 4	Process	CLO2	2	1							1	2	6
Week 5	Thread	CLO2	2	1								2	5
Week 6	Process Scheduling	CLO2	2	1							1	2	6
Week 7	Process Synchronization	CLO2	2	1								2	5
Week 8	MID TERM BREAK												
Week 9	Deadlocks	CLO2, CLO3	2	1							1	2	6
Week 10	Memory Management 1	CLO2, CLO3	2	1								2	5
Week 11	Memory Management 2	CLO2, CLO3	2	1							1	2	6
Week 12	Virtual Memory Management	CLO2, CLO3	2	1								2	5
Week 13	File System	CLO2, CLO3	2	1							1	2	6
Week 14	Implementing File Systems	CLO2, CLO3	2	1								2	5
Week 15	Mass-Storage Structure	CLO2, CLO3	2	1							1	2	6
											SUB-TOTAL SLT :		**77**

Continuous Assessment		%	Face-to-Face (F2F)		Non F2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Lab 1, Lab 2, Lab 3	15	6		2	6	14
2	Quiz 1, Quiz 2, Quiz 3	15	2			4	6
3	Midterm Exam	20	2			4	6
4	Group Project	15				5	5
					SUB-TOTAL SLT :		**31**

Final Assessment		%	Face-to-Face (F2F)		Non F2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Final Exam	35	3			9	12

		SUB-TOTAL SLT :	**12**
		SLT for Assessment :	**43**
		GRAND TOTAL SLT :	**120**
A	% SLT for F2F Physical Component :	45.83	
B	% SLT for Online & Independent Learning Component :	54.17	
C	% SLT for Online Component :	7.50	
D	% SLT for All Practical Component :	0.00	
D1	% SLT for F2F Physical Practical Component :	0.00	
D2	% SLT for F2F Online Practical Component :	0.00	
Please tick (✓) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)			

Identify special requirement or resources to deliver the course (*e.g., software, nursery, computer lab, simulation room etc*)

Virtual Machine Workstation, Oracle Virtual Box, Fedora, Ubuntu

References (*include required and further readings, and should be the most current*)

1. Chakraborty, P. (2023). Operating systems: Evolutionary concepts and modern design principles. Chapman and Hall/CRC.
2. Deitel, H. M. (2003). Operating systems (3rd ed.). Pearson Prentice Hall.
3. Dhamdhere, D. M. (2006). Operating system: A concept-based approach. McGraw-Hill.
4. McHoes, A. M., & Flynn, I. M. (2011). Understanding operating systems (6th ed.). Course Technology, Cengage Learning.
5. Silberschatz, A., Galvin, P. B., & Gagne, G. (2020). Operating system concepts (10th ed.). Wiley.
6. Stallings, W. (2008). Operating systems: Internals and design principles (6th ed.). Prentice Hall.
7. Tanenbaum, A. S. (2006). Operating system: Design and implementation. Prentice Hall.
8. Tanenbaum, A. S. (2023). Modern operating systems (5th ed., Global ed.). Pearson Education Limited.

Academic honesty and plagiarism:

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES)
Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (*if applicable*)

All students must adhere to the following to pass a computing course:
i. Pass both Continuous Assessment AND Final Assessment.
ii. Get at least minimum pass marks following UTM regulation (**40%** or **D+**).

The passing mark for:
- Continuous Assessment is **30%** of continuous marks.
- Final Assessment is **20%** of the final marks.

If a student does not achieve the passing mark for Continuous Assessment OR the passing mark for the Final Assessment, then the student's overall final mark will be set below the UTM passing mark.

Disclaimer:

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While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.

ELT = (Theory + Industrial Guidance + Assessment) x 50%
Total of credit for LI/Practical = ELT/40 Notional Hours

Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by :		**Certified by:**	
Name	: Zuriahati Binti Mohd Yunos, Dr.	**Name**	: Md. Asri Bin Ngadi, Prof. Dr.
Signature	:	**Signature**	:
Date	: 3-Oct-2025	**Date**	: 3-Oct-2025